Mason Davis

noodler-doodler.github.io | linkedin.com/in/masondavis9

EDUCATION

Utah State University – Master of Science in Computer Engineering (GPA: 4.0) Expected July 2026

Utah Valley University – Bachelor of Science in Computer Engineering, Minor in Computer Science (GPA: 3.88) May 2024

EXPERIENCE

Graduate Research Assistant | *Utah State University* August 2024 – Present

- Verified framework correctness for an upcoming FMCAD publication by modeling the VeRAPAk architecture in Dafny
- Ensured Python and Dafny execution parity by engineering a contract-constrained Atheris fuzzer
- Restored adversarial gradient accuracy by resolving a logical mask bug in the ModFGSM implementation

Graduate Teaching Assistant | *Utah State University* August 2024 – Present

- Instructed 15+ students in FPGA design fundamentals by directing digital logic labs utilizing Verilog and AMD Vivado
- Elevated formal verification comprehension by guiding system modeling exercises in Cyber-Physical Systems

Undergraduate Research Assistant | *Utah Valley University* February 2023 – August 2024

- Achieved $>$90% fault identification and localization accuracy by training classification models using TensorFlow
- Accelerated model training by architecting automated computer vision pipelines with OpenCV and CVAT
- Enabled GPS-denied autonomous flight by engineering flight control software featuring real-time YOLO edge inference

Undergraduate Teaching Assistant | *Utah Valley University* August 2023 – May 2024

- Increased student ML proficiency by delivering technical lectures on Support Vector Machines and Deep Learning

PROJECTS

5-Stage DLX Processor | *VHDL, Python, Assembly*

- Maximized instruction throughput by engineering a pipelined VHDL architecture with a Hazard Detection Unit
- Accelerated execution performance at high clock speeds by designing a hardware-based saturating branch predictor
- Enabled UART I/O by integrating Clock Domain Crossing FIFOs and stalling the pipeline for slow operations

Embedded IR-Remote Device Driver | *Verilog, C*

- Achieved hardware-accelerated IR demodulation on a Zybo Z7 FPGA via MicroBlaze processor co-design
- Bridged custom Verilog IP with a Linux application by developing a dedicated C device driver

Tetris | *VHDL*

- Rendered real-time gameplay on a DE10-Lite FPGA by designing a custom VHDL VGA controller
- Validated system logic and timing constraints by performing hardware analysis with Questa and Signal Tap

Gesture-Controlled DJI Tello Drone | *Python, OpenCV, YOLO*

- Enabled gesture-based flight control by training YOLO models for facial authentication and hand tracking
- Minimized command latency by engineering a multithreaded architecture decoupling OpenCV from flight execution

PUBLICATIONS

SPIN (2026), FMCAD(2025), IEEE i-ETC (2024) *Peer Reviewer*

Formal Verification 2026

- Authored an upcoming case study proving termination and correctness of the VeRAPAk verifier in Dafny

Deep Learning 2023 – 2024

- Authored 3 DL papers on fault detection and image classification in *Applied Sciences*, *IEEE i-ETC*, and *Fire*
- Co-authored 2 additional DL papers focused on blade anomaly detection and inspection within *Energies* and *Drones*

SKILLS

Languages: VHDL, Verilog, Python, C, C++, Dafny, MATLAB, Simulink, LaTeX

Tools: Questa, ModelSim, Vivado, Quartus, Cadence, UPPAAL, SpaceEx, SPICE, Multisim, Git, Docker

Libraries & Certifications: TensorFlow, Keras, OpenCV, PyTorch, scikit-learn, FAA Remote Pilot, TensorFlow Developer